Guduri Hasmitha

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.21
Sri Chaitanya Junior College Hyderabad	Class XII (Telangana Board of Intermediate Education)	2024	98.30%
Sree Chaitanya Public School Hanamkonda	Class X (Board of Secondary School Education Telangana)	2022	10.00%

EXPERIENCE

Indian Institute of Technology Indore

Summer Research Intern

- Explored systems programming concepts including **GCC compiler options, ELF binaries,** Linux development tools, and assembly programming through structured assignments.
- Implemented Johnson-Lindenstrauss random projection to reduce sensor feature dimensions from **450 to 50 (88.9% reduction)** for Human Activity Recognition.
- Explored dimensionality reduction techniques for efficient processing of high-dimensional sensor data.

PROJECTS

E-Commerce Analytics & Customer Segmentation Dashboard | *Python, SQL, Power BI*

- Processed and analyzed **392K+** e-commerce transactions using **Python** and **SQL** to identify revenue trends, product performance, and customer purchasing patterns.
- Implemented **RFM** (Recency, Frequency, Monetary) analysis to segment **4,300+ customers** into high-value, loyal, regular, and at-risk groups for customer behavior analysis.
- Developed an interactive **Power BI dashboard** using **DAX**, **KPI cards,** and dynamic filters to visualize revenue, orders, customer segments, and sales performance.

IoT Sensor-based Human Activity Recognition | *C, Linear Algebra*

- Developed a preprocessing pipeline for smartphone sensor data by generating **450-dimensional** motion windows from accelerometer and gyroscope readings.
- Applied **Johnson-Lindenstrauss random projection** to reduce feature dimensionality from **450 to 50 dimensions (88.9% reduction)** while preserving pairwise distances.
- Analyzed the impact of dimensionality reduction on **compression ratio (9:1)** and storage efficiency for high-dimensional sensor data.

GPT-style Decoder-Only Transformer | *Python, PyTorch*

- Developed a GPT-style **decoder-only** Transformer from scratch in **PyTorch** using the TinyStories dataset for autoregressive language modeling.
- Implemented **GPT-2 tokenization**, **multi-head self-attention**, **causal masking**, and a training pipeline for next-token prediction and text generation.
- Built configurable Transformer architectures by varying **context length**, **embedding dimension**, **attention heads**, and **Transformer layers** during model development.

TECHNICAL SKILLS

Data & Analytics: Power BI, SQL (MySQL), Microsoft Excel, NumPy, Pandas, Matplotlib, PyTorch

Programming & Web: Python, C++, C, JavaScript, HTML, CSS

Developer Tools & Systems: Git, GitHub, Linux (Ubuntu), GCC, LLVM/Clang, VS Code

Productivity Tools: Microsoft PowerPoint, Microsoft Word

Core Strengths: Analytical Problem Solving, Strategic Thinking, Communication, Team Collaboration, Presentation Skills

KEY COURSES TAKEN

Computer Science: Introduction to Programming, Data Structures & Applications, DBMS-1, Compilers-1, Algorithms, Theory of Computation

Mathematics: Probability, Linear Algebra, Convex Optimization, Applied Statistics, Calculus, ODE

Other: Linear Systems and Signal Processing, Basic Electrical Engineering

POSITIONS OF RESPONSIBILITY

Chess Event Coordinator | Diesta 2026, IIT Hyderabad

- Coordinated event activities and collaborated with team members to support smooth execution of the chess event.
- Communicated with participants and stakeholders while assisting in event planning and on-ground operations.
- Demonstrated teamwork, organization, and problem-solving in a fast-paced environment.

Social Media Team Member | IKS Student Chapter, IIT Hyderabad

- Collaborated with the team to plan and create social media content for student chapter initiatives and events.
- Planned and executed content strategies to improve audience engagement and maintain consistent communication across digital platforms.

SCHOLASTIC ACHIEVEMENTS

- **IMC Prosperity 4 Trading Challenge 2026:**Secured an Overall Rank of 2366 and an Algorithmic Track Rank of 1602 globally

- **AlgoUniversity Technology Fellowship (ATF) 2025,** Ranked in the Top 8% nationwide among 250,000+ participants

- **Competitive Programming,** Codeforces Pupil (Peak Rating: 1316); actively solved 200+ problems on LeetCode, CodeChef